Bellabeat Smart Device Usage Analysis Data Analysis Case Study

1. Introduction

Bellabeat is a high-tech wellness company that manufactures smart products designed to help women monitor and improve their health habits.
The company collects data related to physical activity, sleep, stress, and other wellness indicators through its smart devices and mobile application.

The Bellabeat marketing analytics team has been asked to analyze smart device usage data to better understand how consumers use smart fitness devices.
The insights from this analysis will help guide future marketing strategies and product positioning.

2. Business Task (Ask Phase)

The objective of this analysis is to examine smart device usage data to identify trends in user behavior.

This analysis aims to answer the following key questions:

1. What trends exist in smart device usage?
2. How could these trends apply to Bellabeat customers?
3. How could these trends help influence Bellabeat's marketing strategy?

The final goal is to generate insights that can support data-driven marketing decisions for Bellabeat.

3. Stakeholders

The primary stakeholders involved in this project include:

Urška Sršen – Co-founder and Chief Creative Officer of Bellabeat.

Sando Mur – Bellabeat co-founder and member of the executive team.

Bellabeat Marketing Analytics Team – Responsible for analyzing user data and generating insights that support marketing strategy.

4. Data Sources (Prepare Phase)

The dataset used in this analysis is the FitBit Fitness Tracker Dataset publicly available on Kaggle.

This dataset contains personal fitness tracker data collected from 30 Fitbit users who consented to share their activity and sleep data.

The dataset includes information related to:
• Daily step counts
• Distance traveled
• Calories burned
• Activity intensity levels
• Sedentary minutes
• Sleep duration
• Time spent in bed

The analysis primarily used the following datasets:
• dailyActivity_merged
• sleepDay_merged

5. Data Preparation (Process Phase)

Before conducting the analysis, the dataset was cleaned and prepared to ensure accuracy and consistency.

The following steps were performed:

- Imported the dataset into Microsoft Excel
- Reviewed the dataset for duplicates and inconsistencies
- Standardized date formats
- Verified column formats and data types
- Combined activity and sleep datasets
- Added helper column (DayofWeek)

6. Data Analysis (Analyze Phase)

Several pivot tables were created to summarize and analyze key metrics such as activity levels, sedentary behavior, sleep patterns, and relationships between activity and calories burned.

6.1 Average Daily Activity by User

This analysis examines the variation in daily activity levels across individual users in the dataset.

The pivot table summarizes the following metrics for each user:

- Average daily step count
- Average sedentary minutes

Observations:

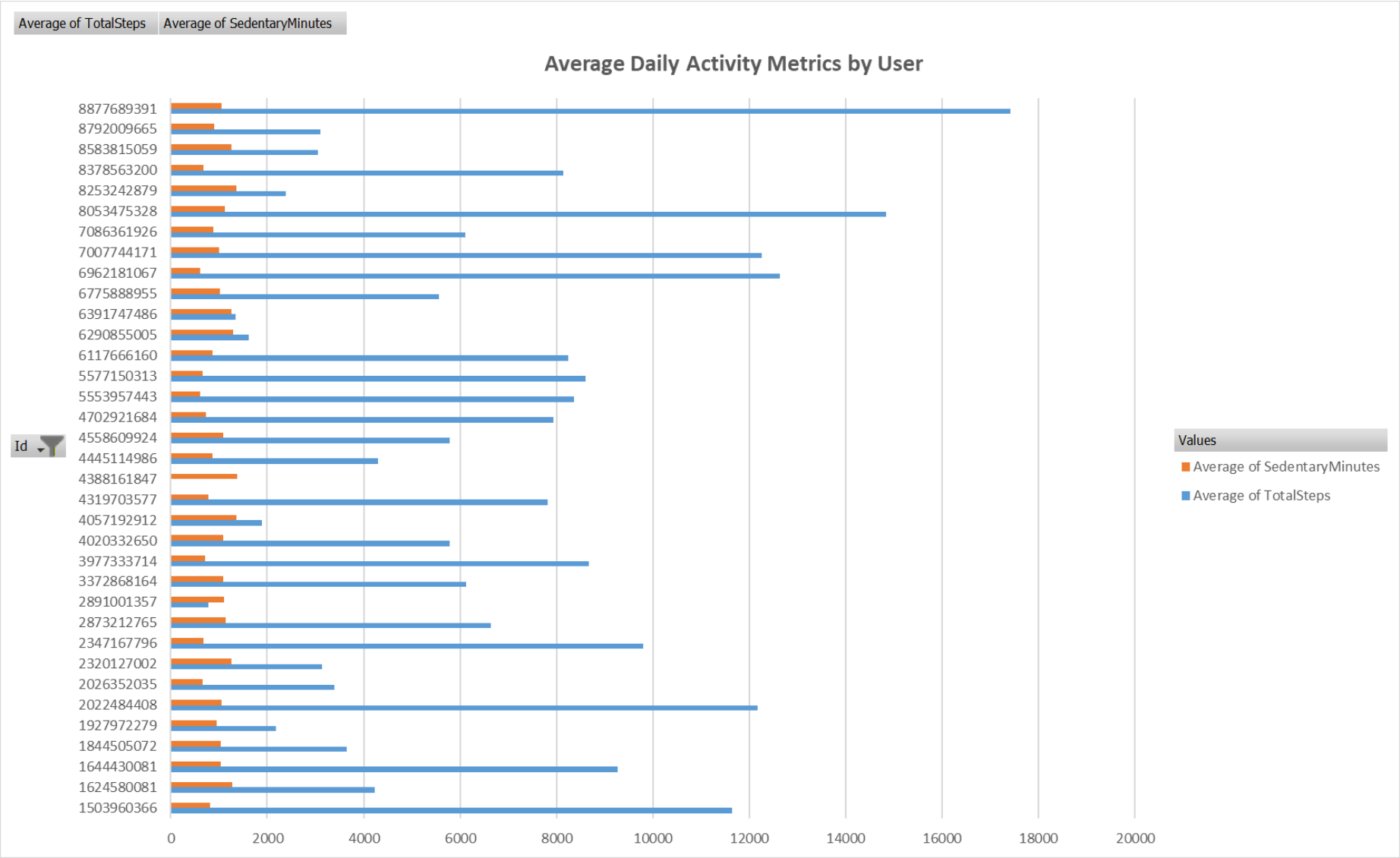
- The results show significant variation in activity levels among users. The average daily step count across all users is approximately 6,546 steps per day, while the average sedentary time is approximately 995 minutes per day.

- A closer examination of the data reveals that only six users record an average daily step count above 10,000 steps, while fifteen users record fewer than 5,000 steps per day. This indicates that a large portion of the users in the dataset maintain relatively low activity levels.

- The dataset also highlights differences in sedentary behavior between users. Some individuals record sedentary minutes below 700 minutes per day, while others exceed 1,300 sedentary minutes.

- Additionally, one user in the dataset recorded zero average daily steps, which may indicate missing data or lack of recorded device activity.

- Overall, the results demonstrate substantial differences in physical activity patterns across users, ranging from highly active individuals to users with predominantly sedentary lifestyles.


Average of TotalSteps
Average of SedentaryMinutes
Average Daily Activity Metrics by User
Id
8877689391
8792009665
8583815059
8378563200
8253242879
8053475328
7086361926
7007744171
6962181067
6775888955
6391747486
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3372868164
2891001357
2873212765
2347167796
2320127002
2026352035
2022484408
1927972279
1844505072
1644430081
1624580081
1503960366
0
2000
4000
6000
8000
10000
12000
14000
16000
18000
20000
Values
Average of SedentaryMinutes
Average of TotalSteps

6.2 Monthly Activity Trends

This analysis examines changes in user activity levels over time by comparing average daily metrics across months.

The pivot table summarizes the following metrics:

- Average daily step count
- Average sedentary minutes

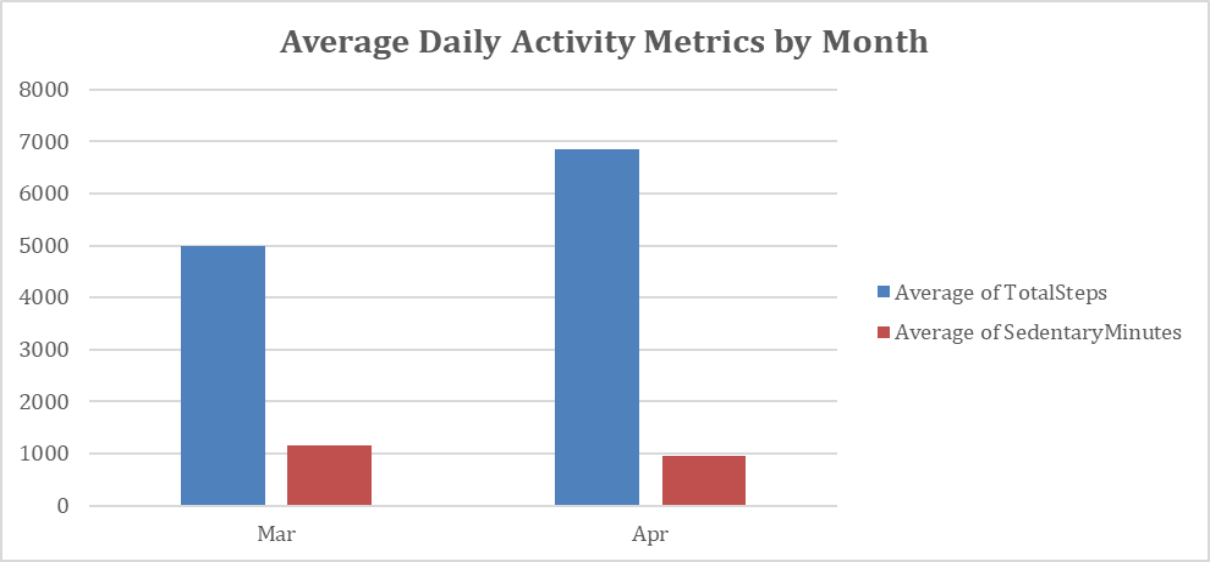
Observations:

- The results show that in March, users recorded an average of approximately 5,001 steps per day, while sedentary time averaged about 1,162 minutes per day.

- In April, the average daily step count increased to approximately 6,845 steps per day, while average sedentary time decreased to approximately 963 minutes per day.

- These results indicate that users recorded higher levels of physical activity and lower sedentary time in April compared to March.

- While the dataset covers a limited time period, the observed difference between the two months suggests that activity levels may vary over time within the dataset.



6.3 Activity Intensity Distribution

This analysis evaluates how users distribute their daily time across different activity intensity levels.

The pivot table summarizes the following activity metrics:

- Very active minutes
- Fairly active minutes
- Lightly active minutes
- Sedentary minutes

Observations:

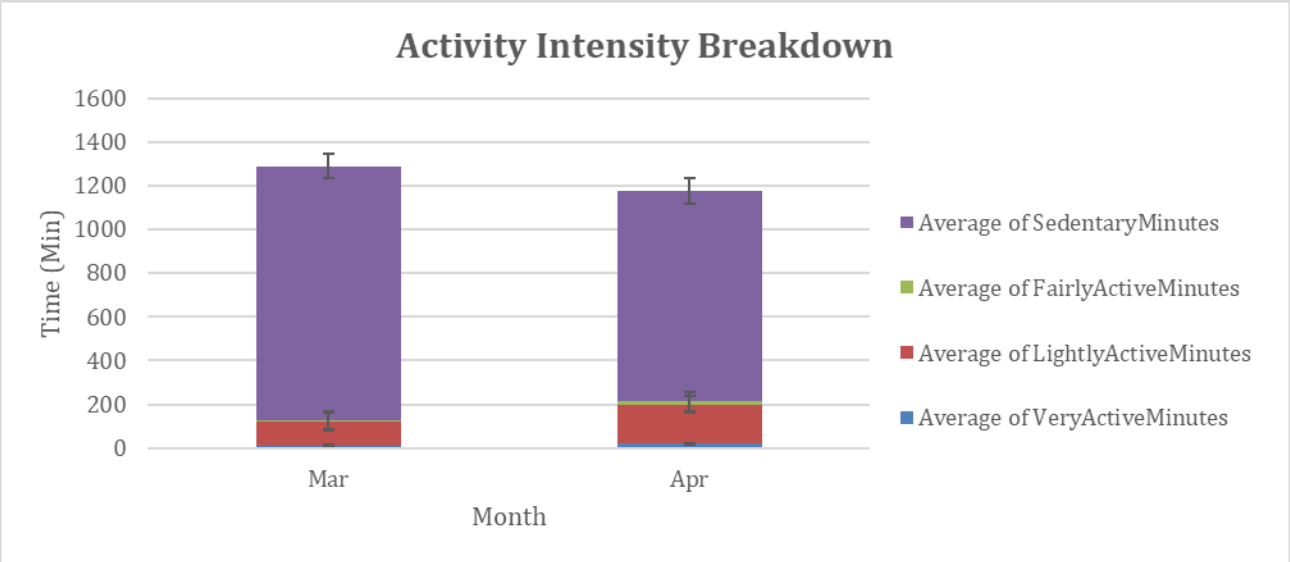
- The results show that sedentary behavior represents the largest portion of recorded time in both months.

- In March, users recorded an average of approximately 1,162 sedentary minutes per day, compared with approximately 113 lightly active minutes, 7 very active minutes, and 7 fairly active minutes.

- In April, sedentary time decreased to approximately 963 minutes per day, while activity levels increased across all activity categories. During this period, users recorded approximately 181 lightly active minutes, 18 very active minutes, and 14 fairly active minutes on average.

- Across both months, light physical activity represents the largest portion of active movement, while moderate and vigorous activity levels remain relatively low.

- Overall, the results indicate that sedentary time dominates daily activity patterns within the dataset, although activity levels increased in April compared to March.



6.4 Sleep Duration and Efficiency Analysis

This analysis examines sleep behavior among users by comparing the average time spent asleep with the total time spent in bed.

The pivot table summarizes two key sleep metrics for each user:

- Average minutes asleep
- Average time spent in bed

Observations:

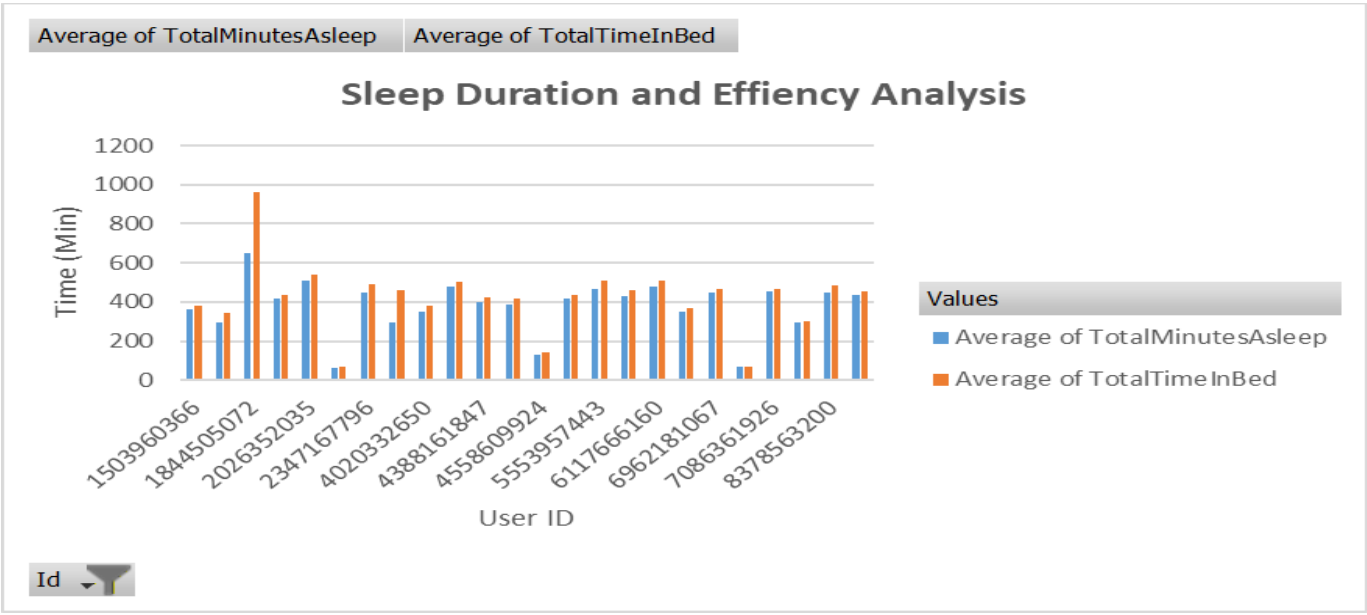
- Across all users, the results show noticeable variation in sleep duration. Some users record relatively long sleep durations, exceeding 450 minutes per night, while others record significantly shorter sleep durations.

- For example, several users record average sleep durations between 430 and 480 minutes, while other users record sleep durations below 150 minutes.

- The analysis also shows that the average time spent in bed is consistently higher than the time spent asleep for all users. This difference represents the period between going to bed and actually sleeping or periods of wakefulness during the night.

- The gap between time in bed and time asleep varies across users. For some users, the difference is relatively small, while for others the difference is larger, indicating variation in sleep patterns within the dataset.

- Overall, the results suggest that sleep duration and time spent in bed differ significantly between users, highlighting diverse sleep behaviors within the dataset.



6.5 Relationship Between Activity and Calories Burned

This analysis explores the relationship between physical activity levels and calories burned by comparing the average number of steps recorded by each user with their average daily calorie expenditure.

The pivot table summarizes the following metrics:

- Average daily step count
- Average calories burned

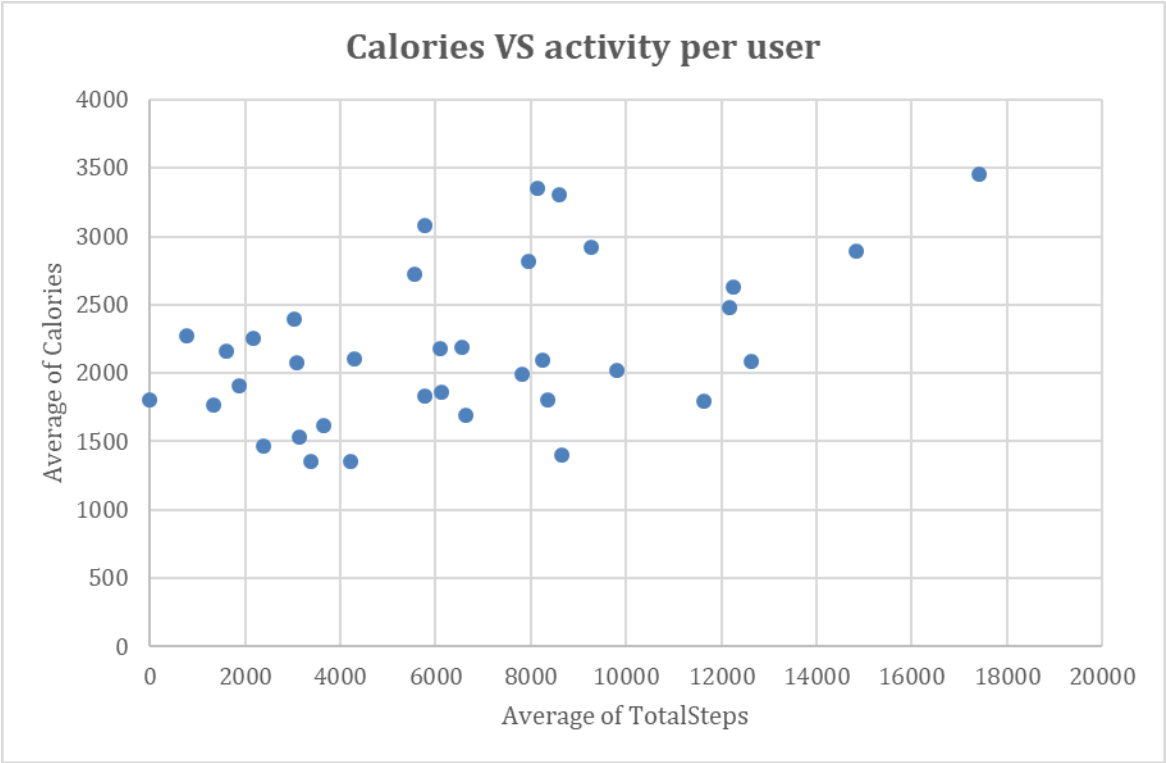
Observations:

- Across the dataset, users with higher step counts generally tend to record higher calorie expenditure. For example, several users who average more than 12,000 steps per day record daily calorie expenditure exceeding 2,000 calories, with the most active users reaching over 3,000 calories burned per day.

- Conversely, users with lower step counts typically show lower calorie expenditure. Several users averaging fewer than 4,000 steps per day record daily calorie values between approximately 1,300 and 1,800 calories.

- However, the dataset also contains a small number of outliers. For example, one user records relatively high calorie expenditure despite very low step counts, and another user records zero steps, which may indicate missing or incomplete activity tracking.

- Overall, the results suggest a general positive relationship between physical activity and calorie expenditure, where increased movement is associated with higher calories burned.



6.6 Activity by Day of Week

This analysis examines how user activity levels vary throughout the week by comparing the average number of steps recorded for each day.

The results show noticeable differences in activity levels across the days of the week.

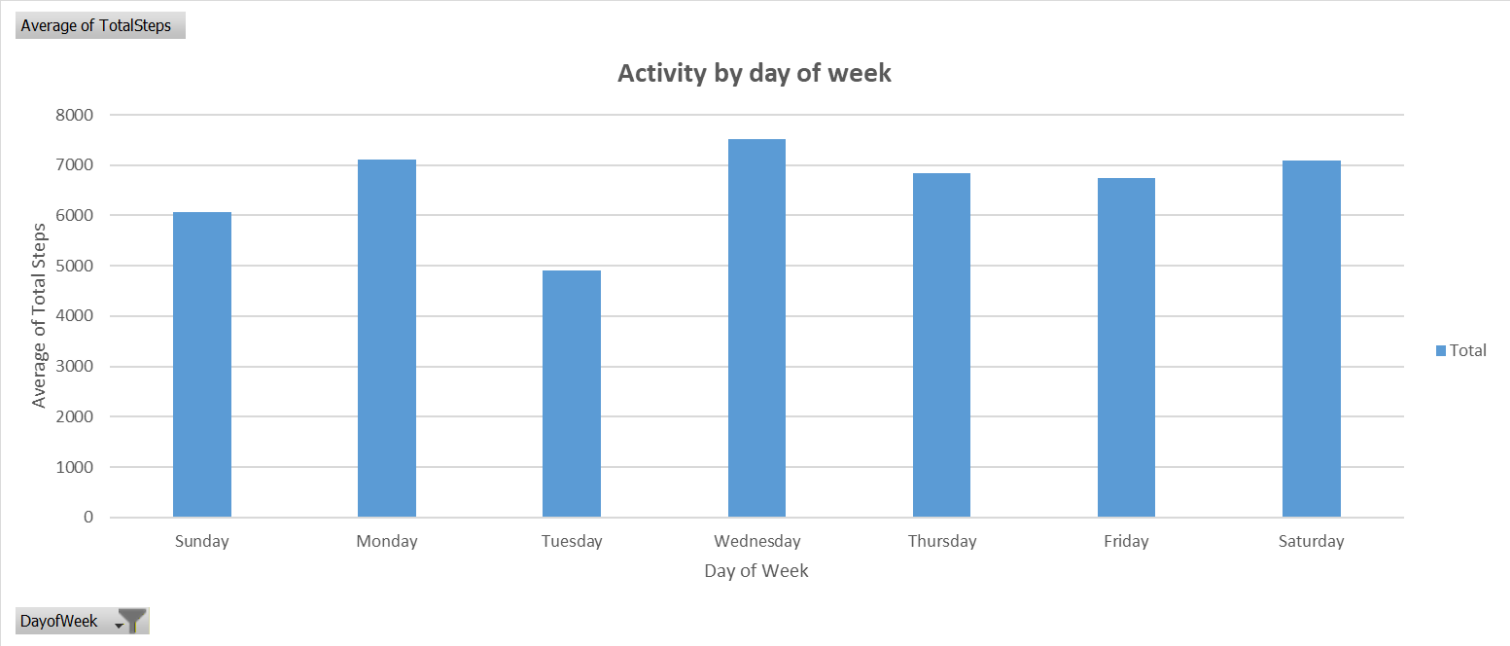
Observations:

- The highest activity levels are recorded on Wednesday, with an average of approximately 7,510 steps per day. Other relatively active days include Monday, Saturday, and Thursday, with step counts ranging between approximately 6,800 and 7,100 steps per day.

- The lowest activity level is observed on Tuesday, where the average step count drops to approximately 4,915 steps per day.

- Overall, the average step count across all days of the week is approximately 6,546 steps per day, which remains below the commonly referenced target of 10,000 steps per day.

- These results indicate that while activity levels fluctuate throughout the week, most users in the dataset maintain moderate activity levels rather than consistently high levels of physical activity.


Average of TotalSteps
Activity by day of week
8000
7000
6000
5000
4000
3000
2000
1000
0
Average of Total Steps
Sunday
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Day of Week
Total
DayofWeek

7. Key Insights

7.1 What are some trends in smart device usage?

The analysis of smart device usage data reveals several trends related to user activity levels, sedentary behavior, and daily movement patterns.

1. Variation in physical activity levels

   The analysis shows substantial variation in daily step counts among users. While some users record more than 10,000 steps per day, many others average fewer than 5,000 steps. This suggests that users of smart fitness devices have diverse activity levels ranging from highly active individuals to relatively sedentary users.

2. High levels of sedentary behavior

   The activity intensity analysis indicates that users spend a large portion of their day in sedentary activity. Sedentary minutes significantly exceed minutes spent in light, moderate, or vigorous activity. This suggests that although users track their physical activity, sedentary behavior still represents a major portion of their daily routines.

3. Fluctuation in activity levels across the week

   Activity levels vary depending on the day of the week. The analysis shows that Wednesday records the highest average step count, while Tuesday records the lowest. This variation indicates that users' physical activity patterns may be influenced by their weekly routines.

4. Relationship between activity and calories burned

   The comparison between step counts and calories burned shows a general trend where users with higher daily step counts tend to record higher calorie expenditure. This indicates that increased movement is associated with higher levels of energy expenditure.

5. Changes in activity levels over time

   The monthly trend analysis shows that average daily steps increased from March to April, while sedentary minutes decreased during the same period. This suggests that user activity patterns may change over time.

7.2 How could these trends apply to Bellabeat customers?

Although the dataset reflects Fitbit users, similar behavioral patterns may exist among Bellabeat customers.

The variation in activity levels suggests that Bellabeat users may also include individuals with different fitness levels and daily movement habits. Some users may already maintain high activity levels, while others may struggle to remain active throughout the day.

The high levels of sedentary behavior observed in the dataset also indicate that many users may spend long periods of time inactive, which highlights an important aspect of everyday lifestyle behavior.

Additionally, the variation in activity levels across different days of the week suggests that user movement patterns may be influenced by work schedules, personal routines, or lifestyle habits.

7.3 How could these trends help influence Bellabeat marketing strategy?

The trends identified in this analysis provide insights that may help guide Bellabeat's marketing strategy.

Understanding that users have varying activity levels allows Bellabeat to target a broad audience with different wellness needs. Marketing messaging could emphasize how Bellabeat products support users at different stages of their fitness journey.

The observation that sedentary behavior occupies a large portion of daily activity suggests that many users may be interested in tools that help them stay active throughout the day.

Finally, recognizing that activity patterns fluctuate across the week may help Bellabeat better understand user behavior and how individuals engage with wellness tracking devices over time.

8. Recommendations (Act Phase)

Based on the analysis of smart device usage trends, several recommendations can be proposed to help Bellabeat improve user engagement and promote healthier activity habits.

Recommendation 1:

Promote Daily Movement Goals: The analysis shows that many users record average daily step counts below the commonly recommended 10,000 steps per day. Bellabeat could encourage users to increase their daily activity levels by promoting step goals and progress tracking within the Bellabeat app.

Marketing campaigns could emphasize how Bellabeat devices help users monitor their daily movement and gradually improve their activity levels.

Recommendation 2:

Introduce Features to Reduce Sedentary Behavior: The activity intensity analysis shows that users spend a large portion of their day in sedentary activity. Bellabeat could introduce or promote features that remind users to move after extended periods of inactivity.

For example, the app could send gentle notifications encouraging users to take short walks, stretch, or stand up during long sedentary periods.

These reminders could help users maintain more consistent movement throughout the day.

Recommendation 3:

Leverage Weekly Activity Patterns: The analysis of activity levels by day of the week shows that user activity varies across different days. Bellabeat could use this information to design personalized activity challenges or reminders tailored to specific days.

For example, the app could encourage users to increase their activity on days when lower activity levels are typically observed.

Recommendation 4:

Highlight Calorie Tracking and Activity Insights: The analysis shows a general relationship between higher step counts and higher calorie expenditure. Bellabeat could highlight this connection in marketing campaigns to demonstrate how increased activity contributes to energy expenditure and overall wellness.

Promoting features that visualize calories burned and activity progress could help motivate users to stay active.

9. Limitations

- The dataset contains information from only 30 users
- The dataset covers a limited time period
- The dataset is most probably outdated (since 2016)
- Some users may have incomplete records or inconsistent device usage

10. Conclusion

This analysis examined smart device usage data to identify patterns in user activity, sedentary behavior, and wellness tracking habits. The results show that users demonstrate varying levels of physical activity, with many individuals recording moderate step counts and significant amounts of sedentary time throughout the day.

The analysis also revealed that activity levels fluctuate across different days of the week and that increased physical activity is generally associated with higher calorie expenditure. Additionally, user activity levels appeared to increase over time within the dataset.

Although the dataset represents a limited number of users and time period, the findings provide useful insights into how individuals interact with smart fitness devices. These patterns can help inform how wellness technology companies, such as Bellabeat, understand user behavior and design products or marketing strategies that support healthier lifestyle habits.